Task 1 | Conception phase

Municipal air-quality readings stored in retryable batches | 17 October 2026

The prototype supports a city dashboard: planners compare hourly readings by station and date, analysts review trends and missing values, and developers query a stable data source. It does not issue health alerts. The sample is UCI's Beijing Multi-Site Air Quality dataset: 420,768 hourly readings from 12 stations, with pollutant and weather measurements. Its size and varied fields represent the municipality's sensor scenario (UCI Machine Learning Repository, n.d.).

MongoDB stores each station-hour as a document with shared fields and a flexible measurements object. This allows new sensor metrics while validating station, timestamp, and source fields (MongoDB, n.d.-a). PostgreSQL would suit a fixed schema and join-heavy use; CSV is easy to exchange but lacks the indexed concurrent query interface planned applications need. An analytical platform may help later, but adds unnecessary setup now.

Deterministic station-time IDs, insert-only upserts, and bounded retries make batch replay safe. Missing values stay absent, not zero. Docker Compose makes the prototype portable. A replica set or sharding may be considered after workload testing; the local container is not highly available, and sharding adds operational complexity (MongoDB, n.d.-b). The plan is to validate the document shape, initialize MongoDB and its station/time index, load batches of 1,000, test reruns, then verify the full container run. Success means complete, non-duplicated station/time queries; misread timestamps or dropped fields are failures.

References: UCI dataset, https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data; MongoDB, Data Modeling, https://www.mongodb.com/docs/manual/data-modeling/; MongoDB, Sharding, https://www.mongodb.com/docs/manual/sharding/.